How to make a soul

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Abstract

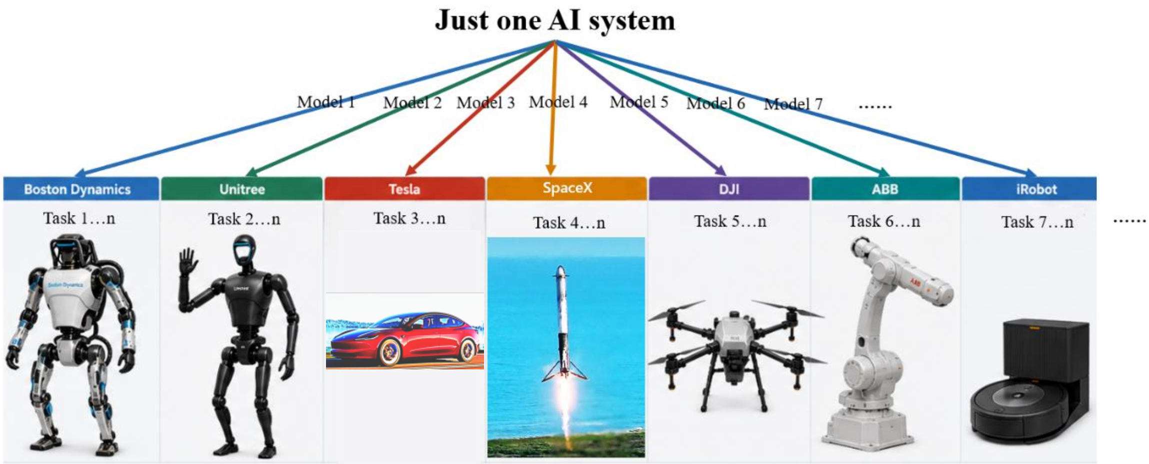

Figure 1.The role of Soul formula

I formulate the Soul:

$$Soul = P(action\ policy \mid world_t^{change}, \text{purpose}, \text{survival})$$

Which not only makes artificial intelligence (AI) be no longer limited to humanity commands to Auto-adapt itself to different machine containers in different fields as shown in Figure 1, but also makes the efficiency of AI who adapts to the physical world be in direct proportion to self-influence at an exponential level (e^x). Furthermore, I am the first to propose (1) the cyber-recognition and cyber-classification of objects based on *Ontology* instead of traditional-rigid classification training based on artificial datasets; (2)world-based changes instead of world as an input into Transformer to model the change patterns for action policy; (3) combination of inter-frame differencing and Transformer at 2026-04-19 19:18:16 UTC+8; (4)Soul Decision Process (V, A, P, S, γ) instead of Markov Decision Process. These will re-standardize the whole AI training process.

[*]All contributions, including conceptual design, research, implementation and manuscript writing, were completed independently by the single author,—Ozymandias Tang.

1 Introduction

At the present stage, most AI researchers focus on areas such as vision [1], hearing, touch, the vestibular system, motor behavior [2] and so on, while the research in core area is lacking,—A phenomenon of animals' reflex arc: Soul， which leads 2 challenges have never been emphasized that makes the efficiency never exceed Human infants for AI to adapt to the physical world:

⑴ AI has never recognized any essence of objects in the world, because it's based on labels of artificial dataset for servicing humanity not for itself to classify objects, which instead reduces the quality of service for humanity.

⑵ AI lacks instinct, who is limited to human commands(as shown in Figure 2), so it is unable to Self-generate behavioral expectations that significantly reduces the efficiency of Reinforcement Learning [3] within World Model [4].

Finally, for the lack of objects recognition and instinct, it's not worth being astonished by emergent phenomena derived just from imitation learning of Transformer [5].

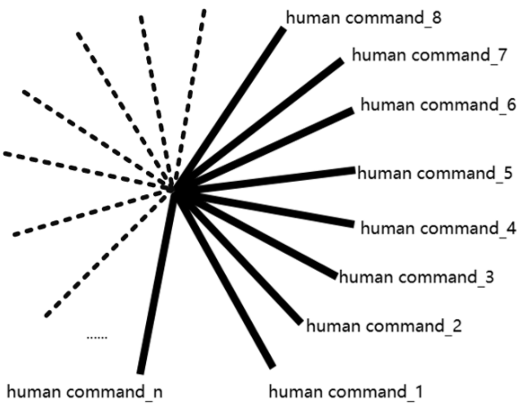

Figure 2.Reinforcement Learning under humanity commands

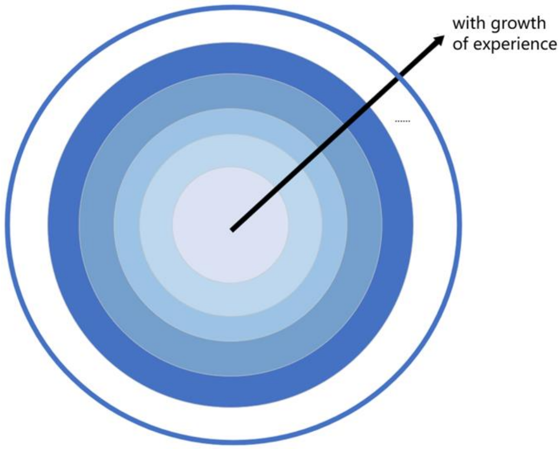

Figure 3.Instinct-driven Reinforcement Learning

Therefore, Soul formula should be the core of artificial intelligence yield, which properly regards the completion of tasks just as a result of instinct rather than the cause that means instinct precedes the tasks, and the completion of tasks is merely a consequence of instinct (as shown in Figure 4). Finally, the efficiency is an exponential level (e^x) for AI to Self-adapt to the physical world, because it is direct proportion to its own state that forms exponential growth system (as shown in Figure 3).

If I define the world as a "vector" and the world changes as a "matrix", that means a matrix acts on a vector, and then both the vector and the matrix must exist simultaneously for the terms to be meaningful. The prior condition of the Soul formula is to detect and utilize the changes of the world. It maps the changes of the world into a global purpose-based survival policy for all different machine containers.

This AI policy will adjust directly the changes of the world to occur in a way that is as beneficial as possible to AI's own survival. Finally, bring the global policy as an implicit standard World Model-based Reinforcement Learning, making the Self-driven learning come true. However, most existing algorithms aren't based on this prior condition of the world change but directly use the temporal features of the raw world as input to Transformer models for sequential modeling, which is unable to validly combine with Soul formula. Therefore, I revise the modeling paradigm of policy algorithms: Only target the changes of the world to feed the high-level abstract features of change values into Transformer models for deep feature learning instead of raw world (as shown in Figure 4), enabling the AI to completely learn the patterns of the world changes for the purpose-based survival policy.

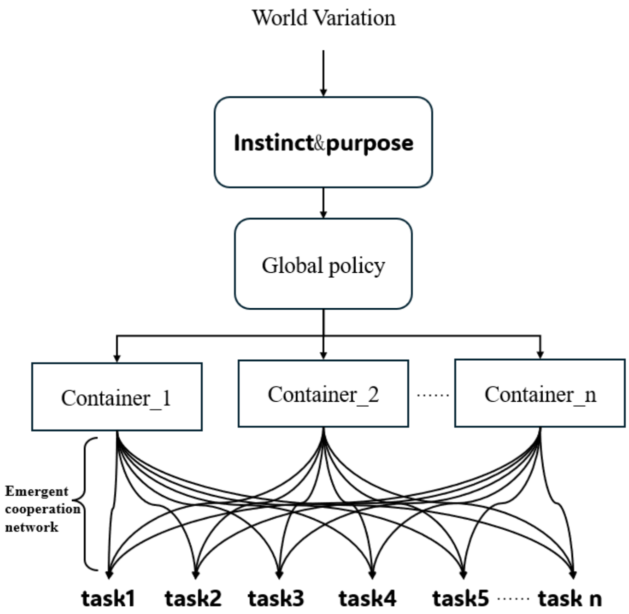

Figure 4.Soul model

In the end, I compare 2 methods for change feature extraction and find a phenomenon: under sufficient data training, the Transformer neural network modeled by the inter-frame difference method [6] can autonomously decouple and learn the expected world change corresponding to multi-dimensional change patterns such as motion, illumination, audio, and environment. This makes up for the shortcoming of the optical flow method [7], which only focuses on the single-dimensional features of motion change and fails to cover the multimodal changes of the complex world such as illumination, background, and occlusion.

2 background

The inter-frame difference method is only used for simple preprocessing tasks such as motion region detection and foreground extraction. It has never been employed as the core extraction of features, nor has it been combined with Transformer to achieve end-to-end inference. Meanwhile, although optical flow method can extract more abundant motion-features than inter-frame difference method, its features are too single-dimensional to cover complex world changes. Thus, the inter-frame difference method remains an indispensable choice in the research of change features, which is suitable for cross-modal scenes in complex scenarios.

3 Soul Formula

3.1 Mother World theory

A mere trend or degree of change lacks any specific-objective semantics, so it cannot be directly prior basis for action policy. Therefore, before extracting the change features, AI must have the ability to recognize all objects and classify objects that should be the core of modeling specific-objective changes. However, “There is no spoon” which citing in movie *The Matrix* [8] means that the physical spoon (beings [9]) has never existed but the prior conceive of it (being [9]). The physical spoon will be changed when its prior conceive changes, which is just as the surficial mode of the existence under the metaphysical existence, — spoon itself. Thus, even Dense, Fine-grained, and Open-vocabulary Classification [10] is still lack of the object-essence-ontology classification, because it’s just based on artificial label-level dataset which just represents the label-level meaning rather than metaphysical-level meaning, resulting in AI never validly recognizing any object to interact in the world.

Therefore, first, I define “Mother Object” that is a result of non-semantically, non-artificially and most-nakedly classifying just via inter-frame pixel change features like humanity who classifies objects according to the integrality, similarity of pixel changes state and the separability of its environmental background. Second, I gift a Survival Instinct to AI. AI can experiment with causing multimodal body change (action) by itself on certain Mother Object to observe its reaction of change. Third, for making the creation more elegant, gift a Purpose to AI that transcends death. Finally, AI can obtain a purpose-based survival coefficient corresponding to self-causing body change, which makes AI sufficiently learn the mapping between self-caused action and the profit of its purpose-based survival.

Furthermore, the purer survival conditions are set directly according to the parameters of its machine container, the less constraints of manual intervention are, the room of mapping is bigger, the lifestyle is more abundant, and finally, the emergent phenomena are more inevitable (Please refer to Section
Emergent Phenomena). Thus, for forming an unbroken semantic space, don't priorly set complex paradigms of training but the core Instinct for posteriorly learning.

In the end, the Mother Object, experiment, Mother Object-based changes and Purpose-based survival coefficient are all embedded in a High-dimensional semantic representation,—"Mother World", which not only generates a tensor of Purpose-Based Survival Instinct Space (as shown in Figure 5) but also births metaphysical conceives: abstract safety, to build up the relationships between self-actions, world-based changes and survival instinct.

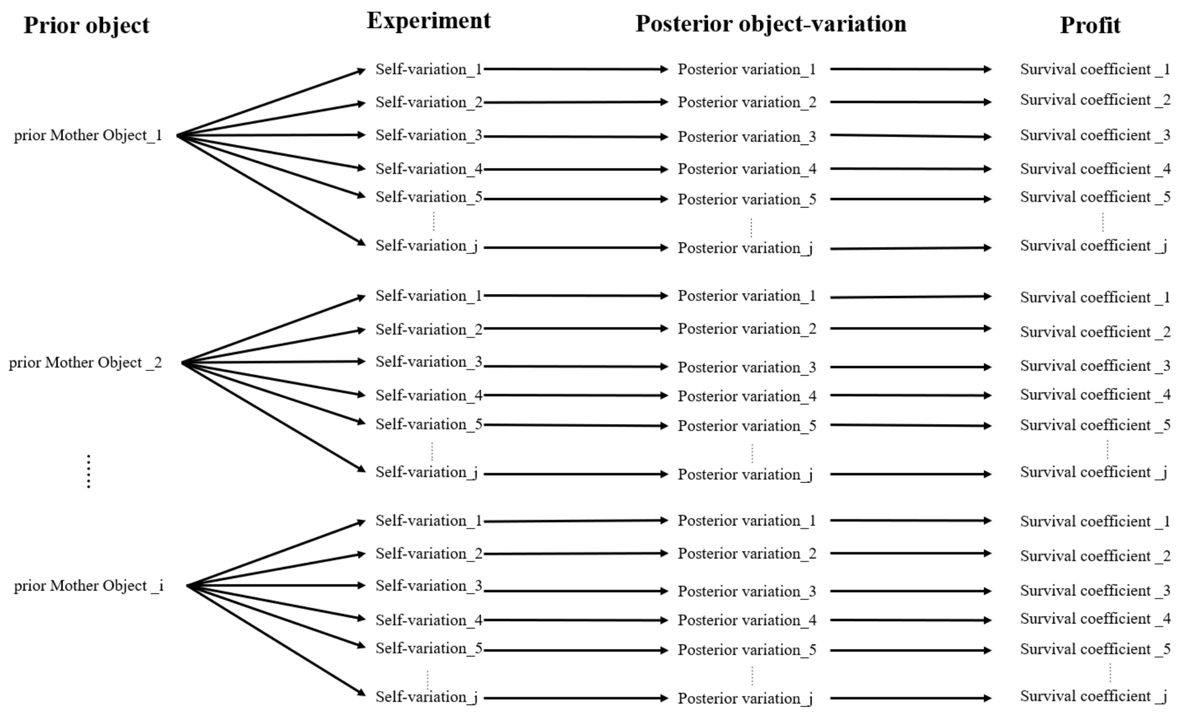

Figure 5.purpose-based Survival Tensor

Only in this way can it be valid for AI to recognize objects rather than classifying them with rigid training of just artificial-formal labels that have been de-abstracted from the Ontological space. In conclusion, Soul formula must follow 3 conditions:

1. Object-Based change features;
2. Purpose;
3. Survival Instinct.

Notably, we only require 3 conditions as inputs into the Transformer model, because under sufficient data training, Transformer-based neural network can autonomously decouple the classes of Mother Objects and implicitly auto-experiment to form a High-dimensional Mother World Tensor, corresponding to multi-dimensional change patterns.

In the end, the implicit recognition of Mother World is shown as Figure 6:

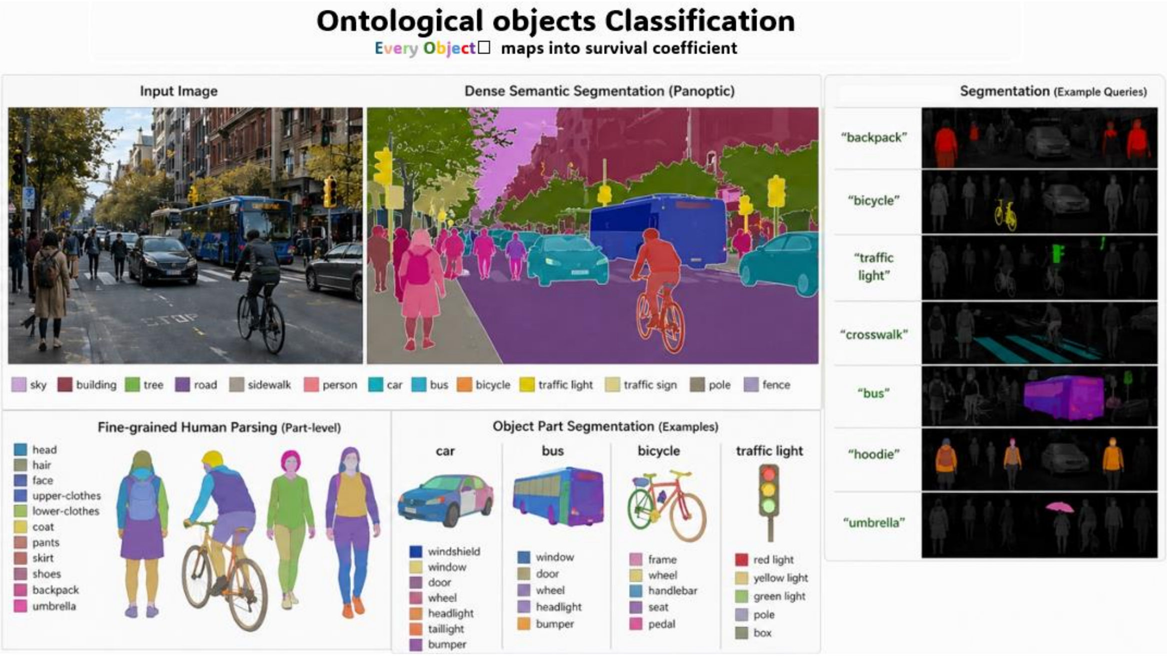

Figure 6.Ontological objects Recognition

3.2 Soul Formula

Because inter-frame difference value is directly equal to the derivative value that directly represents the overall trend and degree of changes and this method can extract multimodal objects changes features with Transformer Model but optical flow method not (Please refer to Section *Extract change*), I use inter-frame difference method in Soul formular.

i. First, define raw temporal frames sequence of the world and inter-frame difference values of this sequence as shown respectively in Equation (1) (2):

$$World_t \in \mathbb{R}^{(T-1)\times C}, t = 2, \dots, T; \qquad (1)$$

$$Change_t := \Delta World_t \in \mathbb{R}^{(T-1)\times C}, \Delta World_t = World_t - World_{t-1}, t = 2, \dots, T, \quad (2)$$

Where C represents the multimodal dimension including visual modality, auditory modality, tactile modality, olfactory modality and so on.

ii. Second, define World-Based changes as shown in Equation (3):

$$World_t^{\mathrm{fusion}} := (World_t, Change_t, World_t \cdot Change_t) \in \mathbb{R}^{3\times(T-1)} \quad (3)$$

iii. third, Linear Fusing. To enhance representation capacity of the interaction, I project the interaction feature into a higher-dimensional embedding space as shown in Equation (4):

$$h_t := W_e\, World_t^{\mathrm{fusion}} + b_e, h_t \in \mathbb{R}^{C\times(T-1)}, \quad (4)$$

where $W_e \in \mathbb{R}^{C\times 3}, b_e \in \mathbb{R}^{C}$ are learnable parameters that map the input fusion into a

$C \times (T-1)$ dimensional space.

iv. fourth, Nonlinear Fusing. The multimodal feature is further fused by a multi-layer perceptron (MLP) to capture higher-order nonlinear interactions as shown in Equation (5):

$$World_t^{change} := \sigma(W_2\, \sigma(W_1 h_t + b_1) + b_2) \in \mathbb{R}^{(T-1)\times C}, \quad (5)$$

where W_1, W_2 and b_1, b_2 denote learnable weights and biases, and $\sigma(\cdot)$ is a nonlinear activation function.

v. Fifth, define survival instincts that are different and flexible owing to the model parameters of the machine container as shown in Equation (6):

$$Survival_i \in \mathbb{R}^{1\times i}, \ (6)$$

Where i denotes the number of the survival conditions to make sure the container is not damaged.

vi. Sixth, to jointly model temporal dynamics World-Based changes sequence and survival standard interactive relationships, I apply a Transformer encoder as shown in Equation (7) with structure-aware self-attention as shown in Equation (8):

$$r_t^{survival} := \text{Transformer}\left(World_t^{change}, Survival_i\right) \in \mathbb{R}^C, \ (7)$$

where:

- $r_t^{survival}$ denotes the contextualized representation;
- Each output token $r_t^{survival}$ corresponds to a unique objective $World_t^{\text{fusion}}$ and $Survival_i$, enabling unified spatio-temporal interaction modeling, under the standard self-attention formulation as shown in Equation (8):

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V. \ (8)$$

This design enables world and survival instinct interaction modeling within a unified attention space. Furthermore, to obtain the final global action policy, I directly aggregate Transformer outputs via a learnable token-wise weighting mechanism as shown in Equation (9):

$$P\left(action\, policy \mid world_t^{change}, \text{survival}\right) = \sigma(W \cdot \sum_{t=1}^{T-1} \frac{e^{w^\top r_t^{survival}}}{\sum_{t'=1}^{T-1} e^{w^\top r_{t'}^{survival}}}\ r_t^{survival} + b) \qquad (9)$$

where:

- $w \in \mathbb{R}^C$ is a learnable importance vector used to adjust the weights of world-based changes for survival action policy,
- $W \in \mathbb{R}^{1\times C},\ b \in \mathbb{R}$ are learnable parameters of the policy head.

vii. In the end, customize a Purpose that defines a rule: The survival must be under the purpose, as shown in Equation (10):

$$P(\text{survival} \mid purpose)\ (10)$$

Final Soul formula is shown in Equation (11):

$$p\left(action\ policy \mid world_t^{change}, \text{purpose}, \text{survival}\right) = \sigma(W \cdot \sum_{t=1}^{T-1} \frac{e^{w^{\top} r_t^{\text{purpose},survival}}}{\sum_{t'=1}^{T-1} e^{w^{\top} r_t^{\text{purpose},survival}}}\ r_t^{,\text{purpose},survival} + b)\ (11)$$

3.3 Soul Decision Process

Traditional RL is modeled as a Markov Decision Process (MDP), defined by the tuple (S, A, P, R, γ). At each timestep, the agent observes a state $s \in$ S, selects an action $a \in A$ according to a policy $\pi(a \mid s)$, receives a reward, and transitions to the next state s' following the environment dynamics $\mathrm{P}(s' \mid s, a)$. The objective is to maximize the expected discounted cumulative reward. However, State and Reward are both confusing in MDP. I re-divide it into 5 stages in Soul Decision Process (V, A, T, S, γ):

1. $world_t^{variation}$: Compare to raw State in MDP
2. Action Policy: the same as MDP
3. State Transition Probability: the same as MDP
4. Survival: Compare it to raw reward in MDP.
5. γ: the same as MDP

Finaly, the overall process is that: ①$world_t^{change}$ comes to AI; ②AI causes the most beneficial $world_t^{change}$; ③The loops of ① and ②.

3.4 Analysis of Efficiency

As AI adapts to more scenarios with more experience, its learning efficiency has been increased in new scenarios, which results in the learning efficiency of AI being directly proportional to its current state, thus exhibiting an exponential growth pattern.

To characterize the self-influenced growth mechanism of the learning system, this study assumes $N(t)$ represents the total number of skills acquired at time r. Since the growth rate of $N(t)$ is proportional to $N(t)$ itself, let the proportionality coefficient be r. Then the system satisfies the differential equation shown in Equation (12):

$$rN(t) = \frac{dN(t)}{dt}\ (12)$$

This equation describes a typical positive feedback mechanism, where the larger the scale of the $N(t)$ system, the faster its growth rate. In the discrete-time case, let the time step be Δt, and the system evolves into $N(t + \Delta t)$. Then, applying a first-order Taylor approximation to $N(t)$ at t, when the time step Δt is sufficiently small, neglecting the higher-order terms $O(\Delta t^2)$, obtain the system evolution as shown in Equation (13):

$$N(t + \Delta t) = N(t) + \frac{dN(t)}{dt}\Delta t + O(\Delta t^2) \approx \lim_{\Delta t \to 0}(1 + r\Delta t)N(t) \quad (13)$$

Therefore, we perform a system evolution on $N(t)$. Let $N(t)$ evolve from the initial time t=0 and denote it as $N(0) = N_0$. Then the final evolution value of $N(t)$ is Equation (14):

$$N(\Delta t) \coloneqq \lim_{\Delta t \to 0}(1 + r\Delta t)N_0$$

$$N(2\Delta t) = \lim_{\Delta t \to 0}(1 + r\Delta t)N(\Delta t) = (1 + r\Delta t)^2 N_0$$

$$\cdots\cdots$$

$$N(n\Delta t) = \lim_{\Delta t \to 0}(1 + r\Delta t)^n N_0 \quad (14)$$

Let $t = n\Delta t$, then the final evolution value of $N(t)$ is given by formula (15):

$$N(t) = \lim_{\Delta t \to 0} N_0(1 + r\Delta t)^{t/\Delta t} \quad (15)$$

Since $\Delta t \to 0$, after the Taylor first-order approximation $\lim\limits_{r\Delta t \to 0} ln(1 + r\Delta t) = r\Delta t$, the final expression for $N(t)$ can be written in exponential form, as shown in Equation (16):

$$N(t) = N_0 e^{rt} \qquad (16)$$

This result demonstrates that the self-driven mechanism inevitably leads to an exponential growth form, where the natural constant e originates from the convergence result of the discrete growth process under the continuous limit. It should be noted that the above exponential growth model is mainly applicable to the initial stage of reinforcement learning. In reinforcement learning, due to factors such as bottleneck constraints caused by environmental changes, the growth process is usually suppressed, and the trend of learning efficiency changes is shown in Figure 7. Therefore, introduce the environmental capacity K to obtain a more general Logistic growth model, as shown in Equation (17):

$$\frac{dN}{dt} = rN(t)\left(1 - \frac{N(t)}{K}\right) \quad (17)$$

1. When $N \ll K$ (i.e., $1 - \frac{N(t)}{K} \approx 1$): Due to a significant skill gap, there is still considerable room for improvement, and learning efficiency grows exponentially.
2. When $N \approx K$ (i.e., $1 - \frac{N(t)}{K} \approx 0$): Since the skill has been largely mastered, further improvement is relatively limited, and the growth rate gradually slows down and eventually stabilizes.
3. When $N > K$ (i.e., $1 - \frac{N(t)}{K} < 0$): Due to a high level of skill proficiency, further improvement is approaching saturation, entering a bottleneck period.

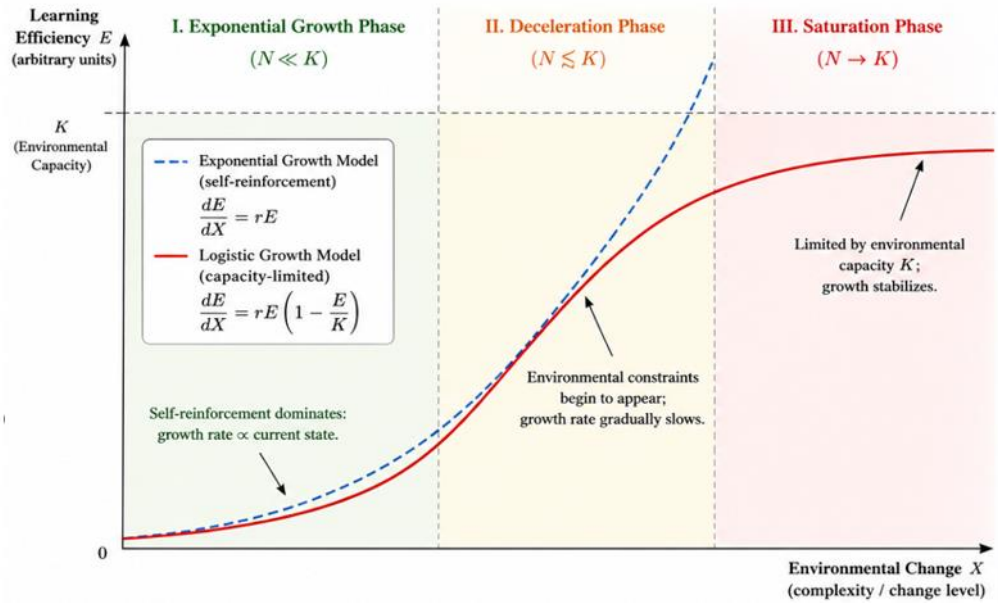

Figure 7.The comparison chart of ideal and actual Learning Efficiency

In summary, Soul formula constructed a positive feedback learning system whose discrete form naturally converges to an exponential function under continuous limits, thus establishing a strict connection between recursive growth and exponential systems.

4 Emergent Phenomena

The soul and emotion are humanity's arrogance toward me who is creator. Axiomatic assumption of soul formula is that: I only set "survival" as the objective, humanity's ultimate goal is nothing more than the emergence of ways to survive.

Excessive research of luxuriant paradigms about training process which squeezes the mapping space of the “Mother World” instead hinders Emergent Phenomena severely. Therefore, the simpler setting of training is, the more quantity of unknown Emergent Phenomena will appear inevitably. Furthermore, I proposed 2 known Emergent Phenomena which implies these 2 following studies *4.1 Cooperation, 4.2 Depth estimation* and *4.3 Causations* may introduce negative effects for emergent phenomena.

4.1 Cooperation

Do not explicitly research the approaches of cooperation between the different containers. For a better purpose-based survival policy, the guidance of cooperation will emerge naturally from global AI system that has been shown in Figure 4. The unique condition is sufficient experience of training.

4.2 Depth estimation

The two camera sensors will produce two images, at the same time, under the constraint of maximized rewards, neural networks will model the relationship between two images to survive that is a Laten 3D space, which means explicit research of 3D reconstruction is impure or imperfect.

4.3 Causation

Do not explicitly research the relationships between future and past for action policy, because the causation never exists [11] but probability [12] that is briefly explained in Figure 8. Therefore, the unique condition of emerging causations whose essence is probability is just sufficient experience of training based on pure purpose-based survival instinct for AI.

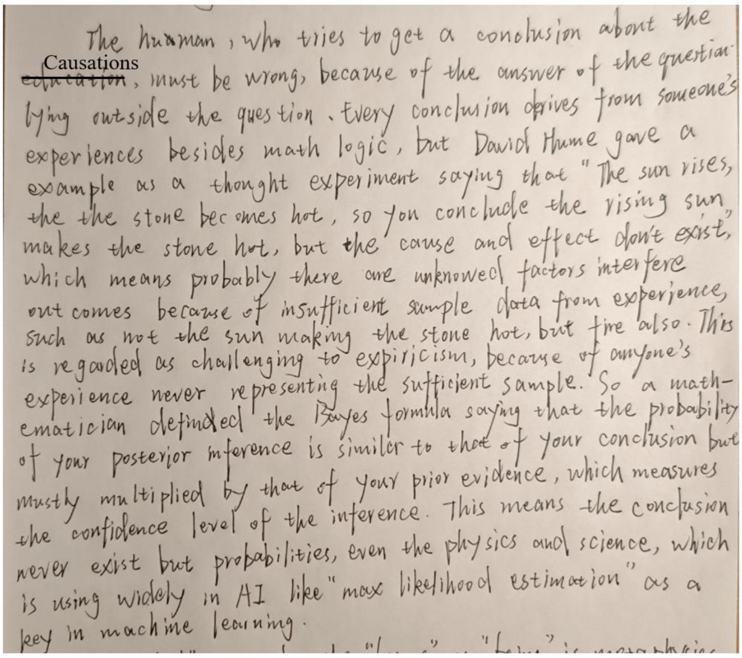
The human, who tries to get a conclusion about the
~~education~~ Causations, must be wrong, because of the answer of the question
lying outside the question. Every conclusion derives from someone's
experiences besides math logic, but David Hume gave a
example as a thought experiment saying that "The sun rises,
the the stone becomes hot, so you conclude the rising sun
makes the stone hot, but the cause and effect don't exist",
which means probably there are unknowed factors interfere
out comes because of insufficient sample data from experience,
such as not the sun making the stone hot, but fire also. This
is regarded as challenging to expiricism, because of anyone's
experience never representing the sufficient sample. So a math-
ematician definded the Bayes formula saying that the probability
of your posterior inference is similer to that of your conclusion but
mustly multiplied by that of your prior evidence, which measures
the confidence level of the inference. This means the conclusion
never exist but probabilities, even the physics and science, which
is using widely in AI like "max likelihood estimation" as a
key in machine learning.

Figure 8.My brief summary of both *A Treatise of Human Nature (David Hume)* and *An Essay towards Solving a Problem in the Doctrine of Chances (Thomas Bayes)*

5 Extract change

There are generally two methods for extracting change features: the inter-frame difference method and the optical flow method. This section compares the advantages and disadvantages of these two methods.

5.1 Optical Flow

1. The basic assumption and mathematical derivation of Optical Flow

The optical flow method is based on the brightness constancy assumption, which states that the brightness in same physical point remains constant over a short period of time. Its basic form is shown in Equation (18):

$$I(x, y, t) = I(x + udt, y + vdt, t + dt) \qquad (18)$$

The first-order Taylor expansion is performed on the right side, as shown in equation (19):

$$I(x + udt, y + vdt, t + dt) \approx I(x, y, t) + I_x udt + I_y vdt + I_t dt \qquad (19)$$

Eliminating the common term $I(x, y, t)$ and dividing by dt, we obtain the classical optical flow constraint equation, as shown in equation (20):

$$I_x u + I_y v + I_t = 0 \qquad (20)$$

Inside of equation (20), u and v represent the spatial velocity of the pixels, I_x and I_y

represent the gradient of the image in the spatial dimension, and I_t represents the brightness change in the temporal direction. This constraint model implicitly assumes that all pixel changes can be explained by spatial motion (u, v), meaning that brightness changes originate entirely from the movement of pixels in space, without including independent appearance change mechanisms (such as color or illumination changes).
Therefore, this model essentially attributes "world changes" entirely to "spatial displacement". In the field of motion analysis, optical flow elevates pixel change features to motion features, including velocity and direction, far surpassing inter-frame difference methods. However, in the comprehensive domain, especially when considering the impact of illumination factors on survival policy—for example, in algorithms based on traffic light on/off cycles—inter-frame difference method far outperforms optical flow method.

2. The limitations of Optical Flow

Optical flow models only express change based on spatial displacement, i.e., pixel motion in space. However, multimodal changes such as audio, illumination and so on do not belong to geometric displacement, resulting in a lack of spatial representation in optical flow models, making action policy unpredictable. I use a algorithm based on traffic light illumination as an example: The spatial gradient values I_x, I_y are 0, when the brightness of the traffic light changes significantly, but its spatial pixel position remains fixed and does not undergo spatial displacement. Substituting the spatial gradient values (0) into the optical flow constraint equation (20), the optical flow constraint equation degenerates as shown in equation (21):

$$I_t = 0 \quad (21)$$

However, the actual brightness of the traffic light changes significantly, as shown in equation (22):

$$I_t \gg 0 \quad (22)$$

Equations (21) and (22) produce an essential contradiction, causing the optimization problem to lose its feasible solution. Therefore, the mathematical conclusion is that the above equation system constitutes an inconsistent equation system. The fundamental reason why it is unsolvable is that the u, v terms completely disappear due to $I_x = I_y = 0$, while the time change term $I_t \neq 0$, which makes it impossible to satisfy the conditions of constraint. Therefore, there is no velocity field (u, v) that can simultaneously satisfy the constraint of equation (20).

5.2 Frame Differencing

Inter-frame difference (IF) is not limited to motion; it can even extract multi-dimensional change features such as illumination, audio, and environment. Furthermore, Since the step between adjacent frames is 1 (Δframe=1)，the inter-frame difference value is directly equal to the derivative value of the pixels in front frame, as shown in Equation (23), representing the overall trend and degree of changes for whole pixels in front frame:

$$f'(frame) \approx \frac{f(frame+\Delta frame)-f(frame)}{\Delta frame} = \frac{f(frame+1)-f(frame)}{1} = f(frame+1)-f(frame) \quad (23)$$

When IF is combined with Transformer, the temporal sequence of pixel derivative values is discretized and mapped into continuous vectors. A multi-layer self-attention mechanism is used to model the global contextual information dependencies in the sequence, allowing the association structure between different tokens to be explicitly expressed in a high-dimensional representation space, forming independent feature clusters in the high-dimensional semantic vector space. This achieves decoupling and multimodal fusion, separating motion from audio and distinguishing between valid and invalid features. It preserves the independence of features in each dimension while achieving synergy of multi-dimensional information, ultimately forming a high-dimensional semantic vector that comprehensively covers multiple variables of the scene, such as motion, lighting, and environment. This compensates for the inaccuracy of motion features in IF compared to optical flow.

5.3 Summary

The advantages and disadvantages of inter-frame difference and optical flow methods are compared as shown in Table 1:

Table 1.the comparison of Optical Flow and Frame Differencing

Comparison Dimensions	Optical Flow	Inter-Frame Difference
Inter-frame Change features	Velocity	Derivative Value
Displacement Change Domain	Stronger	Weaker
Multimodal Change Domain	Weaker	Stronger
Real-time Performance	Weaker (solving the equation $I_x u + I_y v + I_t = 0$ for each frame)	Stronger
Computational Complexity	Higher	Lower

6 Comparative experiment: Optical Flow and Inter-Frame Differencing

6.1 Experimental Setup

The experiments were conducted on the publicly available benchmark behavior recognition dataset Penn Action [13], which contains 15 categories of human actions covering typical scenarios such as sports, fitness, and daily activities, with a total of 2323 annotated samples. The model was implemented and trained using the PyTorch deep learning framework on the Windows operating system with NVIDIA GPUs.

To ensure the controllability and fairness of the comparative experiments, the inter-frame difference method and the optical flow method shared the same Transformer architecture and training strategy. Specifically, the input features were projected into the same latent space dimension ($d_{model} = 512$) through linear mapping. The model employed a 3-layer Transformer encoder, an 8-head multi-head attention mechanism, and a feedforward network with a hidden dimension of 1024. In addition, Dropout was set to 0.1 to alleviate overfitting. For optimization, the Adam optimizer was adopted with a learning rate of 1×10^{-4}, a batch size of 4, and 10 training epochs. The ReLU activation function and CrossEntropyLoss function were used during training.

It should be noted that although the model architecture remained consistent, the optical flow method contains two-dimensional motion information, namely the horizontal and vertical flow components, resulting in differences in the original dimensions of the two input feature types (approximately 512 dimensions for the inter-frame difference method and approximately 1024 dimensions for the optical flow method). Consequently, their statistical distributions and information complexities are not entirely consistent, which further affects the optimization stability and generalization performance of the model during practical training.

6.2 Experimental Results

1. The comparison of Validation Accuracy

Validation accuracy measures a model's ability to generalize on unseen data, as shown in Figure 9. The curves reveal that the validation accuracy of both models shows a continuous upward trend with increasing training epochs, gradually stabilizing, indicating that the models can effectively learn discriminative temporal feature representations. In comparison, the inter-frame difference method converges faster in the early stages and reaches higher accuracy earlier, while the optical flow method shows a relatively slower improvement, resulting in a significant performance difference.

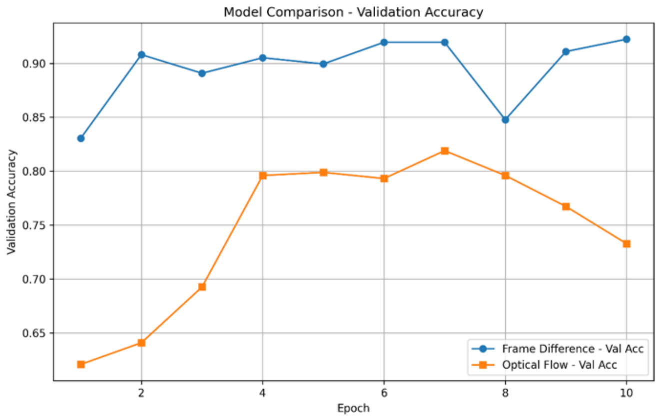

Figure 9. the comparison chart of Validation Accuracy

2. The comparison of Training Loss

The training loss curve reflects the optimization of the model during training, as shown in Figure 10. Both models show a rapid decrease in loss in the initial stage, followed by a stable convergence phase, indicating that the optimization process is stable and gradient descent is effective. In contrast, the inter-frame differencing model shows a faster decrease in loss and reaches a lower level in fewer iterations, indicating that its feature space is easier for the model to optimize and fit.

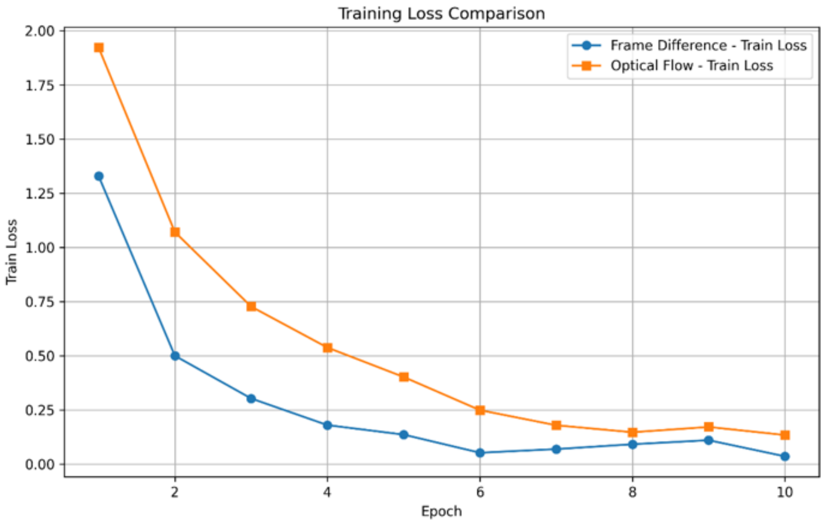

Figure 10.the comparison chart of Training Loss

2. The comparison of Validation Loss

Validation loss measures the magnitude of the model's error on data outside the test distribution, as shown in Figure 11. The results show that both models exhibit a gradual decrease and a tendency to stabilize. Specifically, the inter-frame differencing model consistently maintains a lower validation loss throughout the training process, indicating that its generalization ability is superior to the optical flow model. It also demonstrates smaller prediction errors on unseen data and exhibits more stable temporal modeling capabilities.

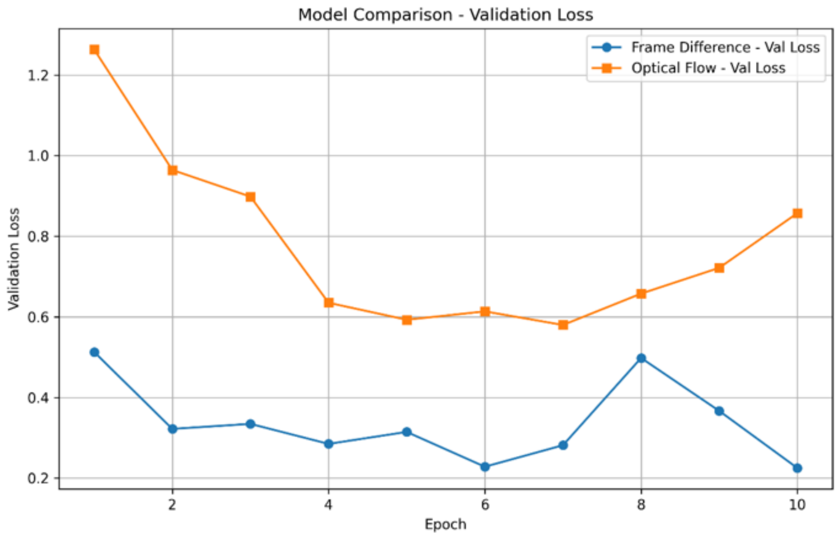

Figure 11.the comparison chart of Validation Loss

6.3 Analysis of Results

A comprehensive analysis of training loss, validation loss, and validation accuracy reveals that the Transformer-based algorithm model exhibits a stable convergence trend under both input feature types. The inter-frame difference method outperforms the optical flow method in both convergence speed and final performance, as evidenced by lower training and validation losses and higher validation accuracy. The three curves show consistent performance: the inter-frame difference method exhibits faster decline, less fluctuation, and better convergence values, with a more rapid and higher accuracy improvement, indicating a more stable optimization process and stronger generalization ability.

From a representation learning perspective, inter-frame difference features provide explicit change signals, enabling the processing of multimodal changes in the world, allowing the Transformer to form a stable attention distribution and accelerate convergence. Simultaneously, its lower feature dimension (512 dimensions) reduces optimization complexity. In contrast, optical flow features have a higher dimension (approximately 1024 dimensions) and a more complex distribution, but are limited to displacement changes, leading to more oscillations during training, slower convergence, and greater fluctuations in validation performance.

In summary, inter-frame difference features are more suitable for Transformer temporal modeling due to their advantages of low dimensionality and multimodality, while optical flow extracts rich motion change features but has a single variable, resulting in relatively weak overall performance.

Conclusion

Although it needs extremely massive training scale for Soul formula to cover every single scenario, the training result is globally shared in this one AI system.

Although it will exhibit a poor performance during the early stage of training, the space of emergence is infinite and global. Inevitably, the global AI system will evolve like humanity from ape to man but at an exponential rate.

The sperm of artificial general intelligence (AGI) has been constructed, all that's missing is a uterus: Super Data Center. Therefore, later generations are only required to construct the uterus. Before the uterus was formed, all AI robots were bubbles.

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